Name_____ Date_____ Period_____

LESSON 2-6 · HOMEWORK PACKET · SHEET 1 OF 2

Median and Outliers

Maryland Standard 6.SP.B.5d · Learning target – I can explain why the median is a better measure of center than the mean when a data set has an outlier.

Sentence Frame: "I know this because..."

REMEMBER

- An **outlier** is a value much greater or much less than the rest of the data.
- An outlier pulls the **mean** toward it. The **median** barely moves, so it is **resistant** to outliers.
- With an outlier, the median usually describes a **typical value** better. Still report the outlier; it is real data.

WORKED EXAMPLE

Six bike rides lasted 14, 18, 15, 16, 64, and 17 minutes. The mean is 24 minutes. Which measure better describes a typical ride?

Step 1 Order: 14, 15, 16, 17, 18, 64. The outlier is 64.

Step 2 Median: (16 + 17) ÷ 2 = 16.5. The mean, 24, is greater than 5 of the 6 rides.

Answer **The median, 16.5 minutes**

PRACTICE

show your work · label units

1. Find the median of each data set. Circle the outlier in each set.

a. 9, 11, 10, 12, 13, 8, 48 _____

b. 3, 70, 72, 75, 78, 71, 74 _____

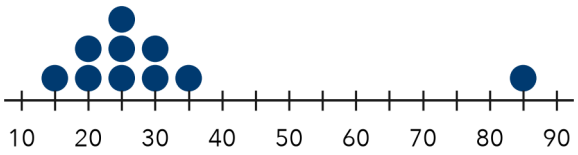
c. 25, 28, 30, 31, 33, 95 _____

d. 40, 41, 43, 44, 46, 47, 150 _____

e. 6, 7, 7, 8, 9, 10, 34 _____

f. 56, 52, 55, 54, 9, 53, 57, 58 _____

2. **Homework Minutes.** The dot plot shows minutes 10 students spent on homework one night. The mean is 31 minutes.



Minutes on Homework

a. Outlier: _____

b. Median: _____ min

c. Students below the mean: _____

d. Better center: _____

3. **Library Visitors.** Visitors on 7 days: 64, 58, 61, 66, 59, 63, and 4 (the library closed early for snow).

a. Median of all 7 days: _____

b. Median without the 4: _____

4. **Car Wash.** A club earned $10, $12, $13, $14, $15, $16, and $60 on 7 Saturdays.

a. Mean: $ ____

b. Median: $ ____

c. Better center: _____

5. **Find the Error.** Andre's data are 31, 33, 34, 36, 37, 39, and 210. He says, "The outlier 210 makes the median much bigger, so the median is not a good measure." Is he right? Find the median to support your answer.

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask someone at home about a normal trip and their slowest trip. Which trip is an outlier?

Name_____ Date_____ Period_____

LESSON 2-6 · HOMEWORK PACKET · SHEET 2 OF 2

Comparing Distributions

Maryland Standard 6.SP.B.5d · Learning target – I can compare two distributions using the median and the interquartile range together.

REMEMBER

- Compare **centers** with the medians. Compare **spreads** with the IQRs (IQR = Q3 − Q1).
- A smaller IQR means more **consistent** data. A full comparison names **both** measures for **both** groups, with numbers.

WORKED EXAMPLE

Team Red: median 14 points, IQR 6. Team Blue: median 18 points, IQR 15. Compare the teams.

Step 1 Centers: 18 > 14, so Blue usually scores more.

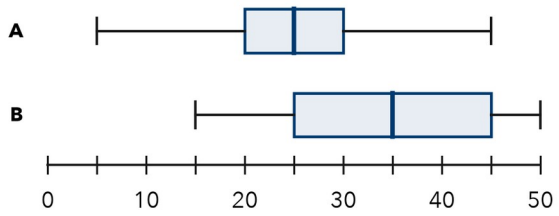
Step 2 Spreads: 6 < 15, so Red is more consistent.

Answer **Blue scores more; Red is more consistent.**

PRACTICE

show your work · label units

1. Reading Minutes. The box plots show minutes read per day by Class A and Class B.



a. Medians: A ___ B ___

b. IQRs: A ___ B ___

c. Usually reads more: _____

d. More consistent: _____

2. Bus Routes. Minutes late on 7 days. **Route 3:** 2, 3, 5, 6, 8, 9, 11. **Route 8:** 0, 1, 4, 6, 10, 14, 15.

a. Route 3: median ___ IQR ___ Route 8: median ___ IQR ___

b. Which route is more predictable? Why? _____

3. Plant Heights (cm). Sun: 10, 12, 13, 15, 16, 18, 19, 21. **Shade:** 6, 7, 8, 8, 9, 10, 10, 12.

a. Sun: median ___ cm IQR ___ cm **b.** Shade: median ___ cm IQR ___ cm

4. Delivery. Shop K: median 28 min, IQR 8 min. Shop L: median 36 min, IQR 8 min. What is the same?

☐ the same center ☐ the same spread ☐ both center and spread

5. Find the Error. Set E: median 70, IQR 25. Set F: median 64, IQR 6. Jada says, "Set E is higher and more consistent." Explain her error.

_____

_____

TEST PRACTICE

choose the best answer

6. Group 1: median 12 cm, IQR 3 cm. Group 2: median 12 cm, IQR 9 cm. Which is more consistent?

A Group 1 **B** Group 2 **C** Both the same **D** Cannot tell

7. What is the IQR of 3, 5, 6, 8, 10, 12, 13?

A 7 **B** 8 **C** 10 **D** 17

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Tell someone at home why a bus with a small IQR is easier to plan around.